

Test Case Document

Project: Halonyx — Secure Decentralized Messaging
URL Under Test: <https://halonyx.onrender.com>
Document Version: 1.0
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1. Test Case Summary

TC ID	Test Case Name	Requirement	Status
TC-01	Landing Page Load	FR-01	Pass
TC-02	USID Generation via Registration	FR-01	Pass
TC-03	User Login	FR-02	Pass (after fix)
TC-04	Main Dashboard Display	FR-01, FR-05	Pass
TC-05	Send and Receive Encrypted Messages	FR-03, FR-04	Pass
TC-06	Local Message Storage	FR-05	Pass
TC-07	Trigger Emergency Broadcast	—	Pass
TC-08	Dismiss Emergency Alert	—	Pass

2. Detailed Test Cases

TC-01 — Landing Page Load

Field	Details
TC ID	TC-01
Requirement	FR-01
Objective	Verify that the application landing page loads correctly
Preconditions	Application is deployed and accessible
Test Step	Navigate to https://halonyx.onrender.com
Expected Result	Landing page loads with "Create your Secure ID" prompt visible
Actual Result	As Expected
Status	Pass

Field	Details
Notes	—

TC-02 — USID Generation via Registration

Field	Details
TC ID	TC-02
Requirement	FR-01
Objective	Verify that a unique cryptographic Secure ID (USID) is generated upon registration
Preconditions	Landing page is loaded (TC-01 passed)
Test Step	Enter a valid Full Name and Email Address, then submit
Expected Result	A unique decentralized identifier (Secure ID) is generated and displayed
Actual Result	As Expected
Status	 Pass
Notes	—

TC-03 — User Login

Field	Details
TC ID	TC-03
Requirement	FR-02
Objective	Verify that a registered user can log in successfully
Preconditions	User account has been registered (TC-02 passed)
Test Step	Enter login credentials and submit the login form
Expected Result	User is authenticated and redirected to the main dashboard
Actual Result	Initially failed; login succeeded after database schema fix
Status	 Pass (after fix)
Notes	Defect: Login failed on first attempt due to incorrect database schema. Resolved by updating the schema. Re-test passed.

TC-04 — Main Dashboard Display

Field	Details
TC ID	TC-04
Requirement	FR-01, FR-05
Objective	Verify that the dashboard correctly displays user identity and empty contact state
Preconditions	User is logged in (TC-03 passed)
Test Step	Access the main dashboard after login
Expected Result	Dashboard displays the user's unique Secure ID and a "No contacts yet" message
Actual Result	As Expected
Status	 Pass
Notes	—

TC-05 — Send and Receive Encrypted Messages

Field	Details
TC ID	TC-05
Requirement	FR-03, FR-04
Objective	Verify that messages are encrypted, relayed, and received between two users
Preconditions	Two user accounts exist and are logged in; contact added
Test Step	Initiate a chat session between two users; send a message from User A to User B
Expected Result	Message is sent and received successfully by both parties using end-to-end encryption
Actual Result	Successfully Completed
Status	 Pass
Notes	Encryption via Signal Protocol; server relays without storing message content

TC-06 — Local Message Storage

Field	Details
TC ID	TC-06
Requirement	FR-05
Objective	Verify that messages are stored only on the user's local device and not on the server
Preconditions	Messages have been exchanged (TC-05 passed)
Test Step	Inspect server-side storage / database after message exchange
Expected Result	No message content is persisted on the server; messages are accessible only locally
Actual Result	As Expected
Status	 Pass
Notes	Validates the No Persistence non-functional requirement

TC-07 — Trigger Emergency Broadcast

Field	Details
TC ID	TC-07
Requirement	—
Objective	Verify that an emergency broadcast is displayed correctly to the user
Preconditions	User is logged in and on the dashboard
Test Step	Trigger an emergency broadcast from the appropriate control
Expected Result	A red emergency banner and an alert toast notification appear on screen
Actual Result	As Expected
Status	 Pass
Notes	—

TC-08 — Dismiss Emergency Alert

Field	Details
TC ID	TC-08

Field	Details
Requirement	—
Objective	Verify that the emergency broadcast banner can be dismissed by the user
Preconditions	Emergency broadcast is active (TC-07 passed)
Test Step	Click the "Dismiss" button on the emergency broadcast banner
Expected Result	The emergency banner is removed from the UI
Actual Result	As Expected
Status	Pass
Notes	—

3. Defect Log

Defect ID	TC ID	Description	Severity	Resolution	Status
DEF-01	TC-03	User login failed due to incorrect database schema	High	Database schema updated to match application expectations	Resolved

4. Requirement Traceability Matrix (RTM)

Requirement ID	Requirement	Use Case	Design Component	Test Case	Test Status
FR-01	USID Generation	User Registration	USID Module	TC-01, TC-02, TC-04	Pass
FR-02	User Authentication	Login	Auth Module	TC-03	Pass (after fix)
FR-03	Encrypted Messaging	Send Message	Signal Encryption	TC-05	Pass
FR-04	Message Relay	Message Transfer	Node.js Relay	TC-05	Pass
FR-05	Local Storage	View Message	Local Storage	TC-04, TC-06	Pass

5. Test Execution Summary

Total Test Cases Passed Failed Pass After Fix Not Executed Suspended

8 7 0 1 0 0

Overall Result:  **All test cases passed**

One test case (TC-03) required a database schema fix before passing.